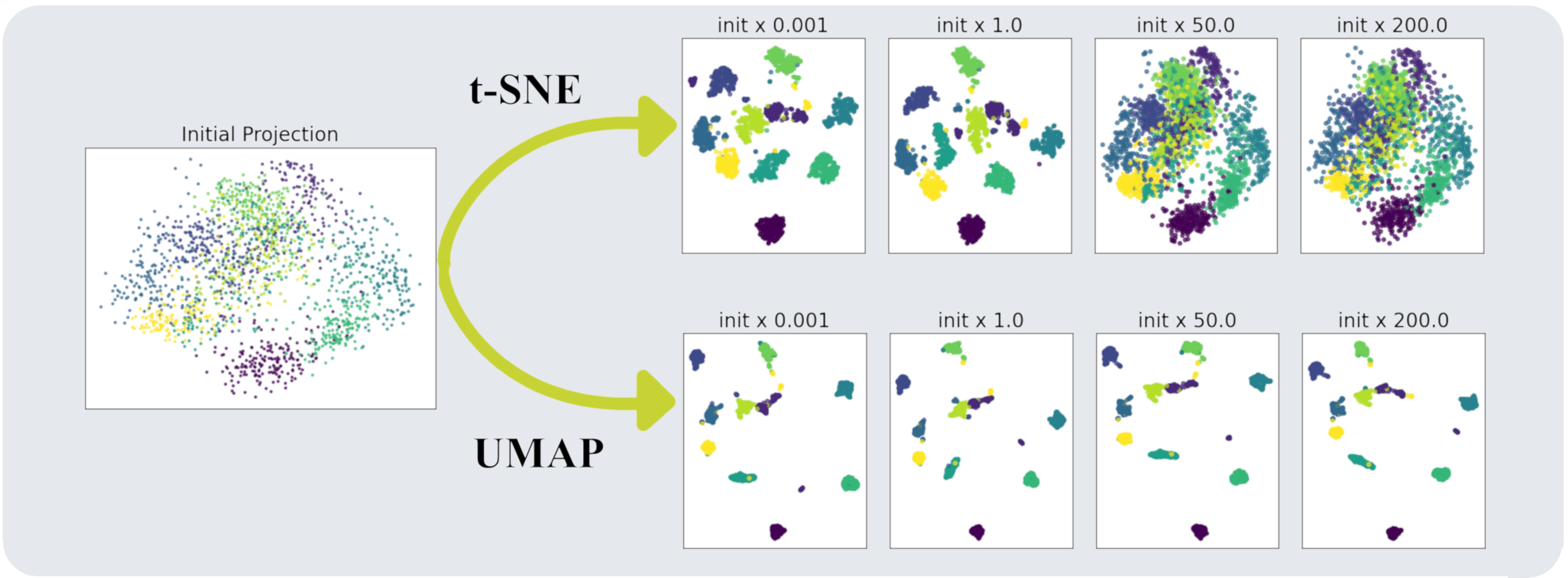
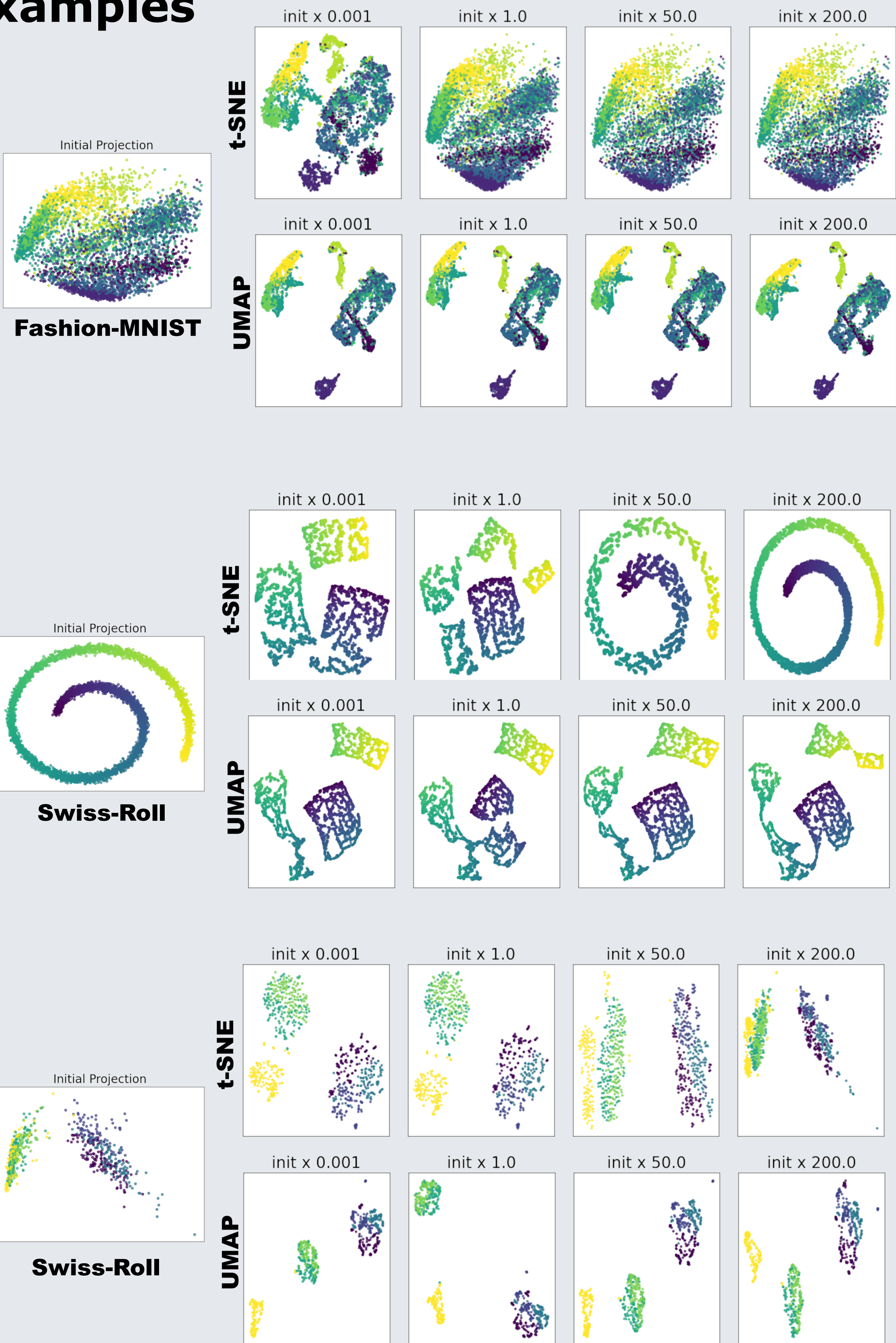


t-SNE is Sensitive to Initialization Size (but UMAP is NOT)

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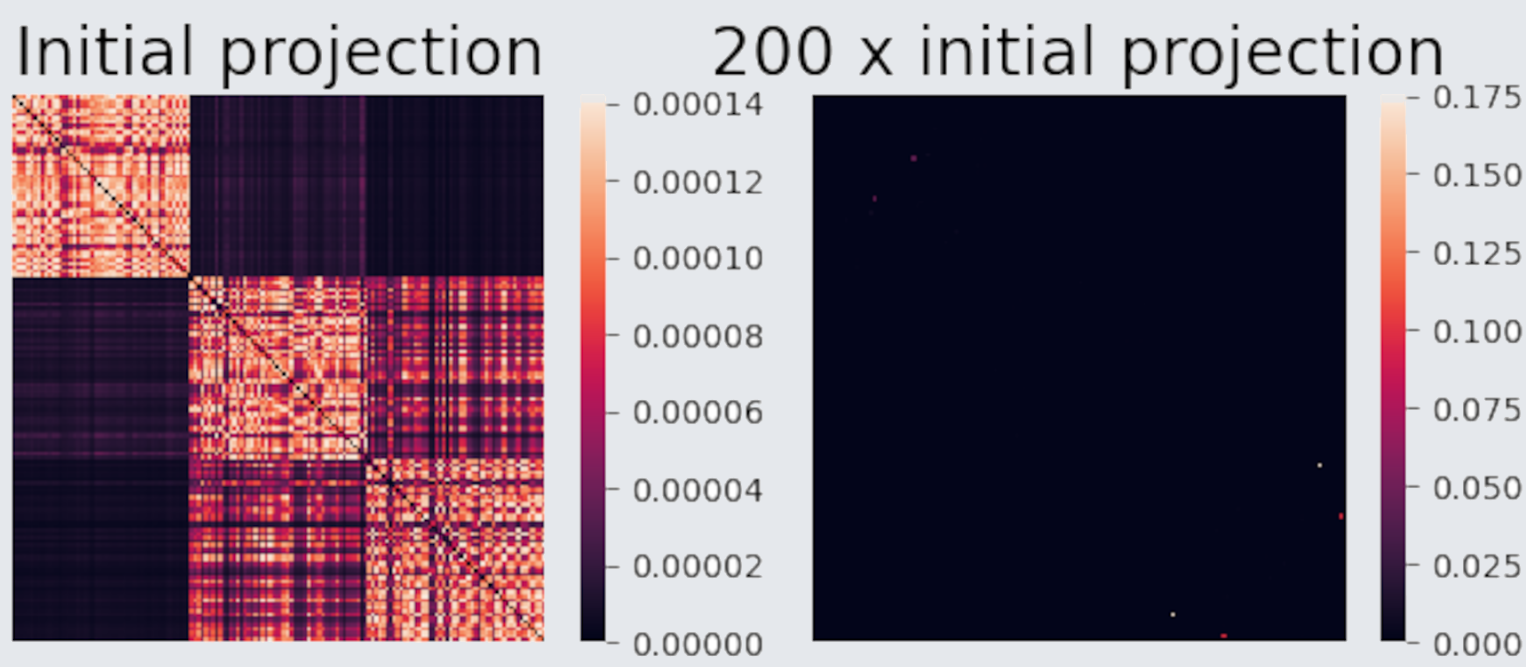


Examples



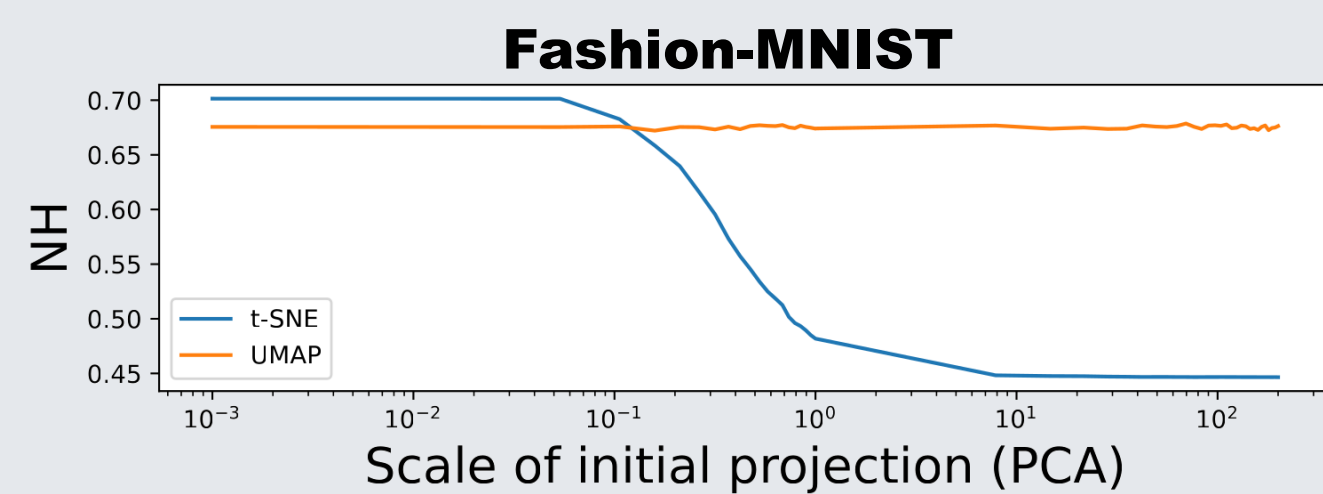
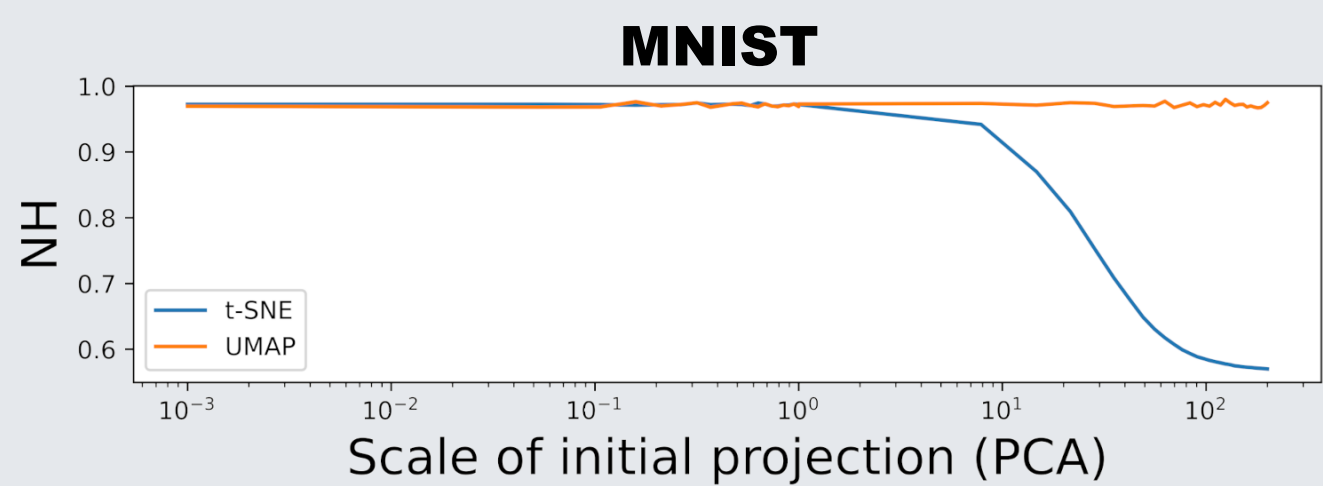
Intuition

- The attractive and repulsive forces are imbalanced
- The constructed similarity matrix (Q) approaches uniformity as the size increases



Evaluation

- Vary the initialization size of t-SNE and UMAP
- Measure the neighborhood hit of resulting embedding
- t-SNE tends to lose quality as size increases



Conclusions

- Don't ignore the size of t-SNE initialization
- For large datasets, scale the initialization to unit width
- Investigate further differences of t-SNE and UMAP

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